

221752

Time : 3 Hrs.

M.M. : 60

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 COP of heat pump is _____ than that of Refrigerator.
a) Less b) More
c) Equal d) None of the above
- Q.2 One Ton of refrigeration is equal to :
a) 3.5 KW b) 6.5 KW
c) 5.5 KW d) 2.5 KW
- Q.3 Which of the following is Halo-Carbon refrigerant?
a) R-11 b) R-500
c) R-718 d) R-170
- Q.4 Effect of superheating of vapour is :
a) High COP b) Low COP
c) Same COP d) None of the above
- Q.5 For saturated air :
a) Wet bulb depression is zero
b) Wet bulb depression is positive

- c) Wet bulb depression is negative
- d) Web bulb depression can either be positive or negative

Q.6 Air refrigeration cycle is used in _____.

- a) Commercial refrigerators
- b) Domestic refrigerators
- c) Gas liquification
- d) Air conditioning

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The value of COP should be greater than _____.

Q.8 An ideal refrigerant should have high boiling point. (True/False)

Q.9 Air without moisture content is called _____.

Q.10 Define sensible heating.

Q.11 The part of refrigeration system, where actual cooling takes place is called _____.

Q.12 Define refrigeration effect.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain Liquid nitrogen refrigeration.

- Q.14 Draw proper T-S and p-h diagrams of simple Vapour Compression Refrigeration Cycle.
- Q.15 Explain Sensible Heating and Humidification processes.
- Q.16 Write a short note on horizontal scaled reciprocating compressor.
- Q.17 Define 1) Halo-carbon compounds
2) Azeotropes
- Q.18 Explain By pass Factor.
- Q.19 Explain thermostatic expansion valve.
- Q.20 Mention the safety devices used in refrigeration system. Explain any one of them.
- Q.21 Explain Reverse Carnot Cycle.
- Q.22 Mention the advantages and disadvantages of Air-Cooled condensers.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the following air conditioning terms:
- a) Latent heat
 - b) Wet bulb temperature
 - c) Degree of saturation
 - d) Dew Point temperature
- Q.24 Write down the properties of following refrigerants:
- a) R-717
 - b) R-22
- Q.25 Explain Auto Defrosting. Write its advantages and disadvantages.

(2360)

(4)

221752